



Important: Text generated by AI and may contain incorrect information, always check the content.
To identify outliers, we will use the Interquartile Range (IQR) method, which is a common technique for detecting outliers in a dataset.

```
```python
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

Check for numerical columns
numerical_columns = df.select_dtypes(include=['int64', 'float64']).columns

Calculate Q1 and Q3 for each numerical column
Q1 = df[numerical_columns].quantile(0.25)
Q3 = df[numerical_columns].quantile(0.75)

Calculate IQR for each numerical column
IQR = Q3 - Q1

Identify outliers using IQR
outliers = ((df[numerical_columns] < (Q1 - 1.5 * IQR)) | (df[numerical_columns] > (Q3 + 1.5 * IQR))).sum()

Display the count of outliers for each column
display('Count of outliers in each column:')
display(outliers)
```

'Count of outliers in each column:'
```

| | 0 |
|--|-----|
| baseline value | 0 |
| accelerations | 14 |
| fetal_movement | 307 |
| uterine_contractions | 1 |
| light_decelerations | 150 |
| severe_decelerations | 7 |
| prolongued_decelerations | 178 |
| abnormal_short_term_variability | 0 |
| mean_value_of_short_term_variability | 70 |
| percentage_of_time_with_abnormal_long_term_variability | 309 |
| mean_value_of_long_term_variability | 71 |
| histogram_width | 0 |
| histogram_min | 0 |
| histogram_max | 24 |
| histogram_number_of_peaks | 19 |
| histogram_number_of_zeroes | 502 |
| histogram_mode | 73 |





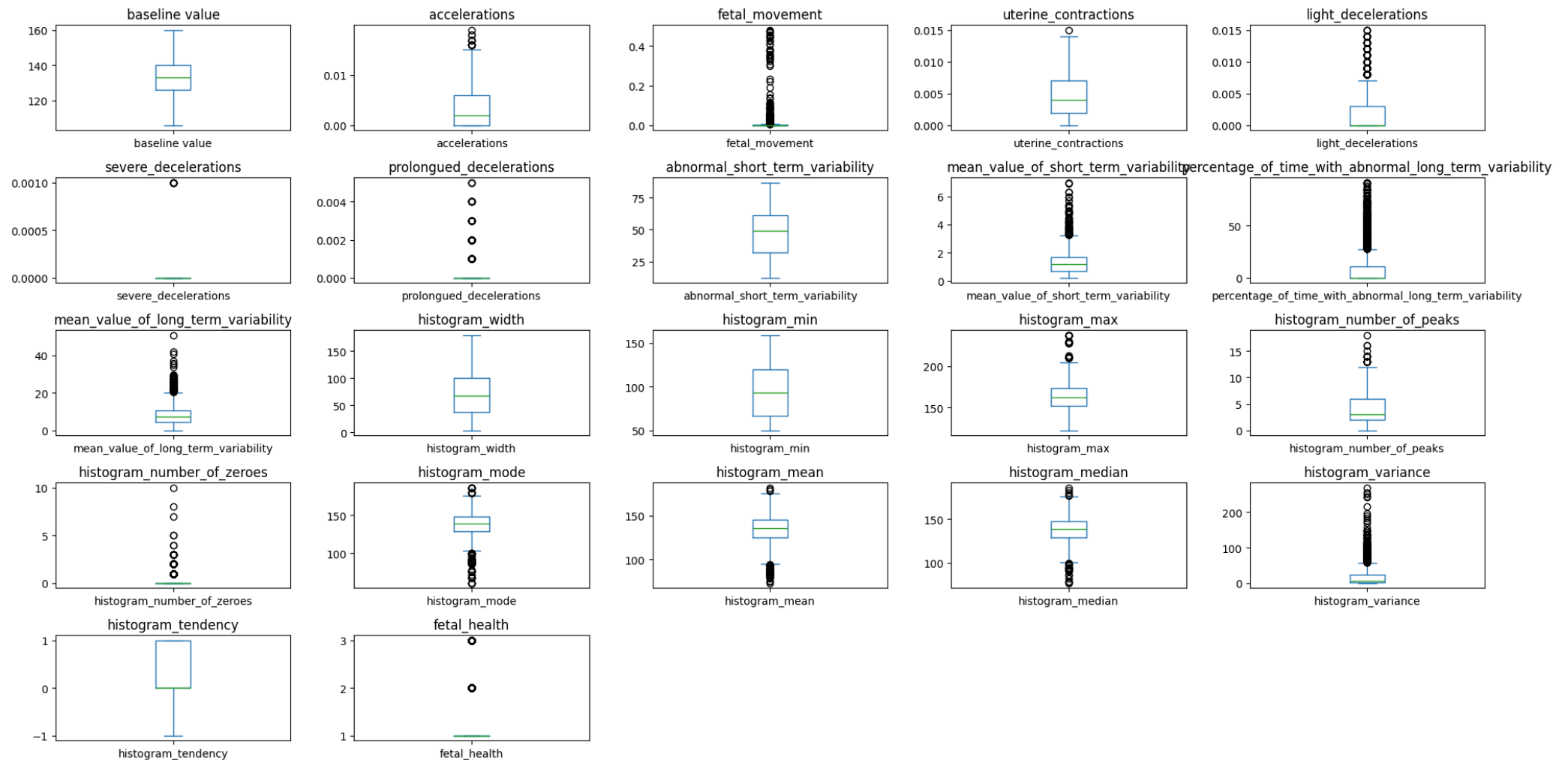
| | 0 |
|--------------------|-----|
| histogram_mean | 45 |
| histogram_median | 28 |
| histogram_variance | 184 |
| histogram_tendency | 0 |
| fetal_health | 471 |

`dtype: int64`
Before creating boxplots to visualize the outliers, we will check the presence of outliers in the data.

The boxplot is a standardized way of displaying the distribution of data. It is a graphical representation that displays the five-number summary: minimum, first quartile, median, third quartile, and maxi

```
```python
Create boxplots for numerical columns to visualize outliers
plt.figure(figsize=(20,10))
for i, col in enumerate(numerical_columns):
 plt.subplot(5, 5, i+1)
 df[col].plot(kind='box', subplots=True, figsize=(20,10))
 plt.title(col)
plt.tight_layout()
plt.show()
```
```





Before performing the above steps, we will check if there are any categorical or null values.

```

python
# Check for categorical columns
categorical_columns = df.select_dtypes(include=['object', 'category']).columns

# Display categorical columns
display('Categorical columns:')
display(categorical_columns)

# Check for null values
null_values = df.isnull().sum()

# Display null values
display('Null values in each column:')
display(null_values)

```

'Categorical columns:'



Index([], dtype='object')
'Null values in each column:'

| | 0 |
|--|---|
| baseline value | 0 |
| accelerations | 0 |
| fetal_movement | 0 |
| uterine_contractions | 0 |
| light_decelerations | 0 |
| severe_decelerations | 0 |
| prolongued_decelerations | 0 |
| abnormal_short_term_variability | 0 |
| mean_value_of_short_term_variability | 0 |
| percentage_of_time_with_abnormal_long_term_variability | 0 |
| mean_value_of_long_term_variability | 0 |
| histogram_width | 0 |
| histogram_min | 0 |
| histogram_max | 0 |
| histogram_number_of_peaks | 0 |
| histogram_number_of_zeroes | 0 |
| histogram_mode | 0 |
| histogram_mean | 0 |
| histogram_median | 0 |
| histogram_variance | 0 |
| histogram_tendency | 0 |
| fetal_health | 0 |

dtype: int64

